

Task	CTC	Mask-mixing gain
BEIR FiQA	$O_T(N)$	+0.011
Outlier (scale- k)	$O_T(NM)$	+0.505
QDmatch (FiQA)	$O_T(N^2)$	+0.031
Reordering [†]	$O_T(N^2)$	+0.182
Contradiction [†]	$O_T(N^2)$	+0.791

Table 1: **What curriculum mask-mixing buys, by corpus-traversal complexity.** Both chunked arms use the identical document-chunked mask at evaluation time and differ only in training: *+Mix* anneals a per-step unrestricted-attention coin $p_{\text{standard}}: 0.8 \rightarrow 0$, *Chunked* never sees an unrestricted step. *Gain* is $(+Mix) - (Chunked)$. Qwen3.5-4B, identical shards, identical ladders, matched hyperparameters; *eval_size* = 500 per rung, so a gain under ~ 0.04 is within noise. Each figure is the mean per-rung gain over the 2k–32k ladder (5 rungs; reordering stops at 16k, 4 rungs). Mixing is worth nothing on $O_T(N)$ retrieval (FiQA, +0.011) and is the difference between learning the task and not learning it at all on contradiction (+0.791) and outlier detection (+0.505). *Reordering and QDmatch earn their entire gain at the shallow rungs*: by the deepest rung the full-attention arm has itself fallen below 0.15, leaving no headroom for either chunked arm, so their gains run +0.473 and -0.073 at 2k but +0.000 and +0.006 at the bottom of the ladder. [†]Pure-chunked training cross-entropy never descended on these two tasks (contradiction: $1.175 \rightarrow 1.071$, flat across the whole run), while FiQA and QDmatch on the identical recipe converged normally – so the collapse is task-specific rather than an infrastructure failure, but it reflects a failure to fit the training set and is not by itself a statement about what the mask can represent.